Week 2

COGS 108 Wi 2025

Due dates

- D2: Due on Sunday (Jan 26th)
- Q3: Due on Sunday (Jan 26th)
- PLEASE ACCEPT YOUR GITHUB INVITATION
    - Check your mailbox, accept invitation
    - This is necessary for your group project

Datahub/JupyterLab

Do not forget to submit(time-stamps)

Submitted assignments		
Practice_COGS108_FA24	COGS108_FA24_A00	Fetch Feedback
2024-09-28 07:25:28.062723 UTC		
2024-10-02 23:16:02.164214 UTC		
2024-10-04 05:57:15.640169 UTC		
PythonReview_COGS108_FA24	COGS108_FA24_A00	Fetch Feedback
2024-10-01 02:44:25.028562 UTC		

Important things to note

- Validate != Submit
- might need to optimize your code if runs too long
- Do not re-submit after due dates
- Do not delete the original cell

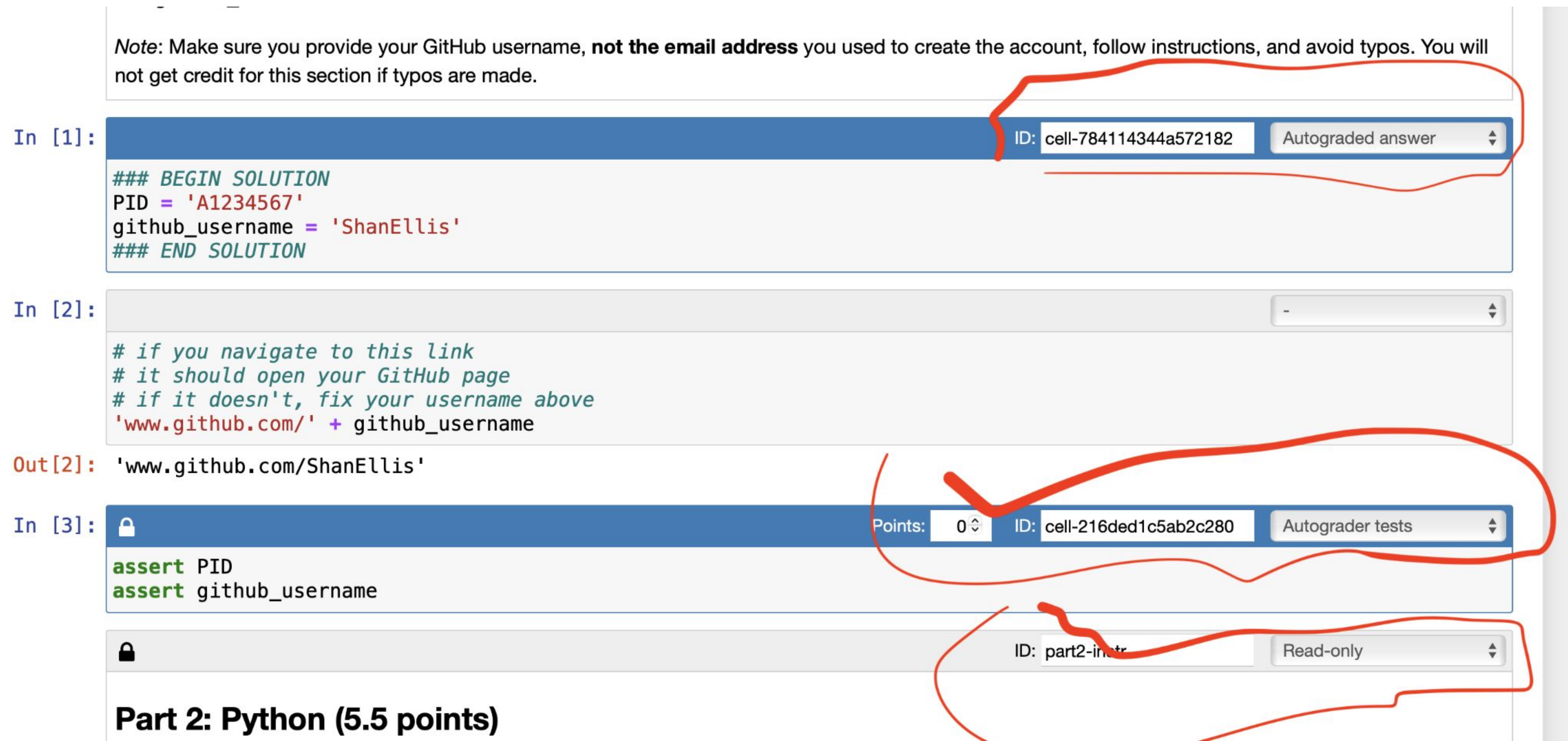
Datahub/JupyterLab

Do not re-submit after due dates

- Please double check before submitting
- JupyterLab won't store your code

Do not delete the original cell

- autograde based on cell ID



Github

1. Fork

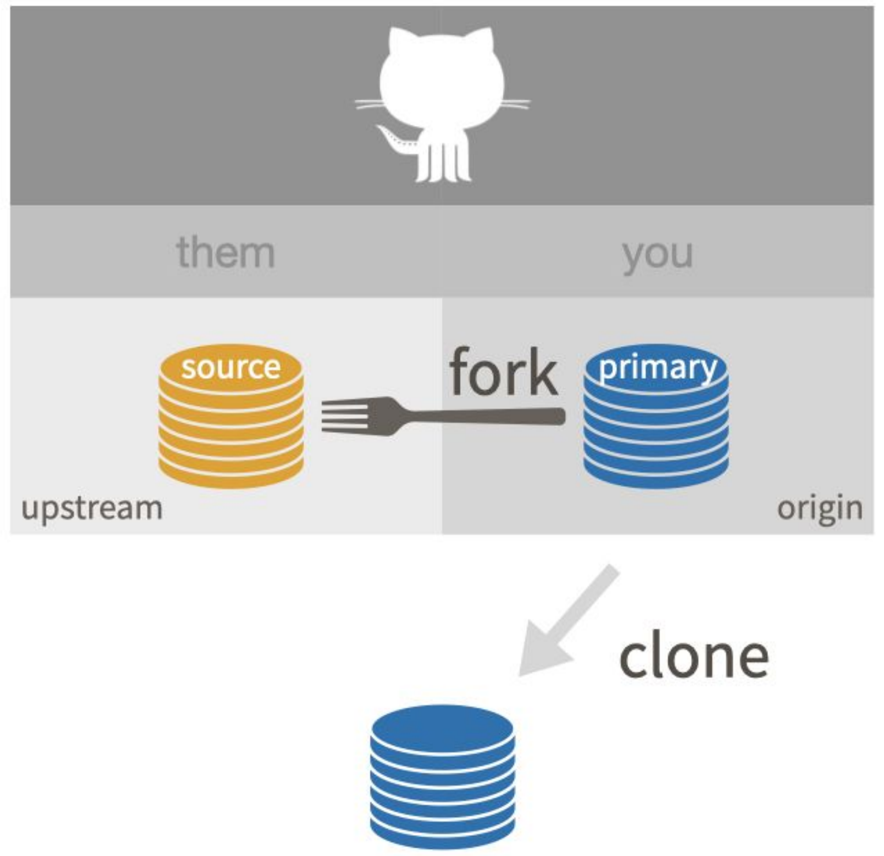
   A “fork” is just an independent copy of a repository that you can develop on without affecting the original.

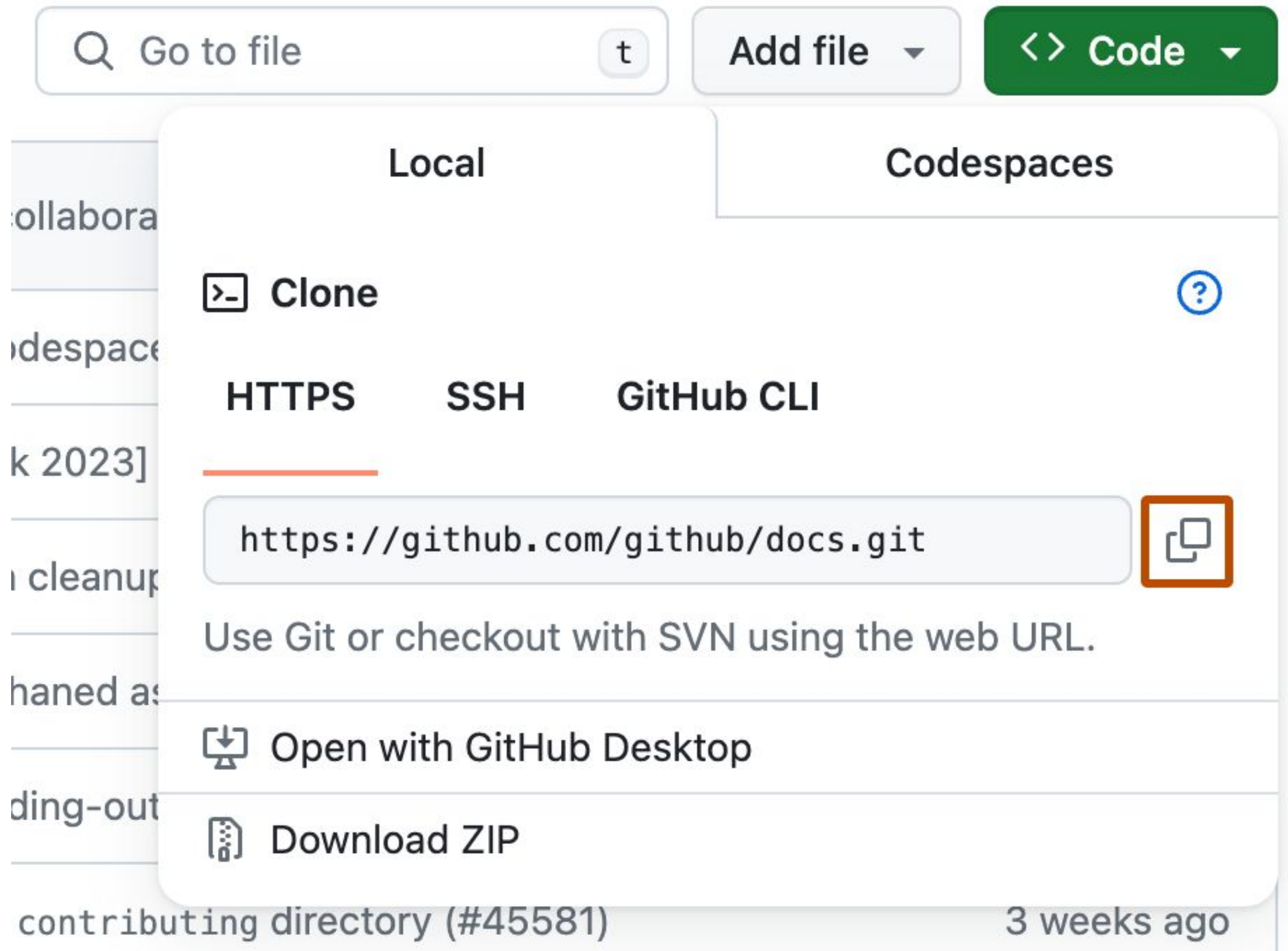
2. Clone

   Creates a local copy of a repository on your machine.

3. You can “push” changes back to the remote repository if you have permissions.

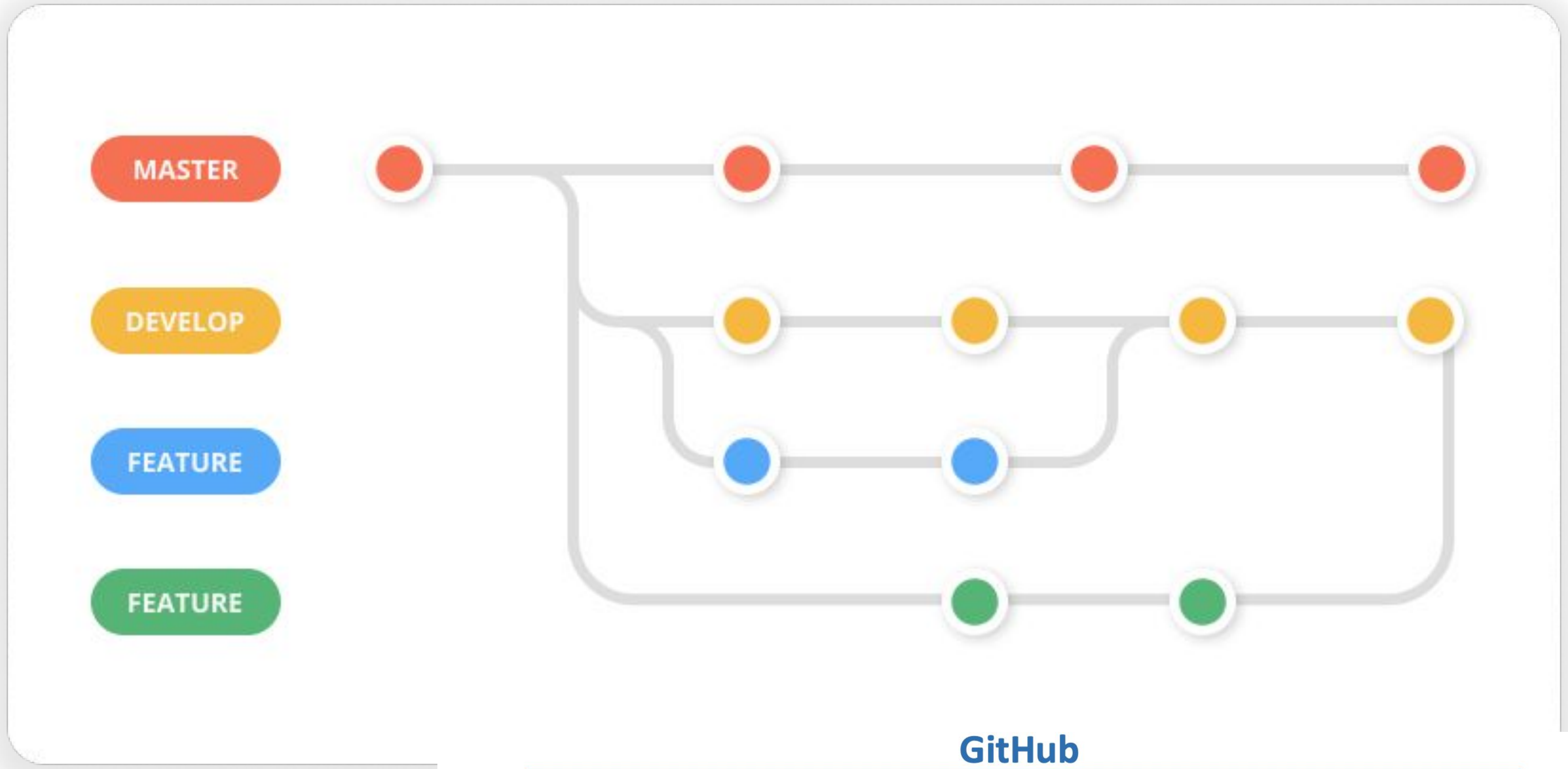
$ git clone https://github.com/user/repo.git





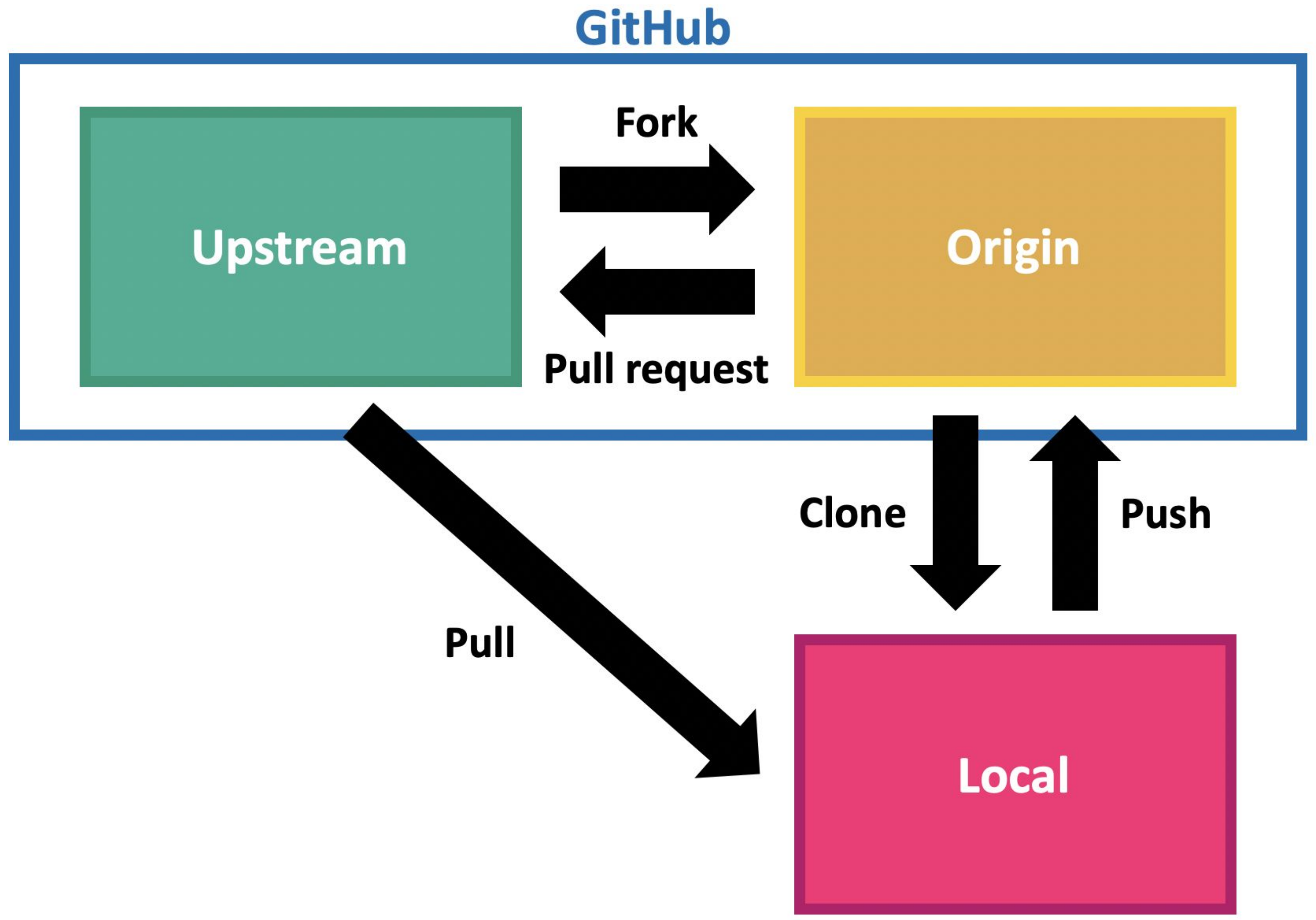
Branching

Branch will be independent of the clean, functioning "master" code and is a safe place for you to delete, modify and add code.



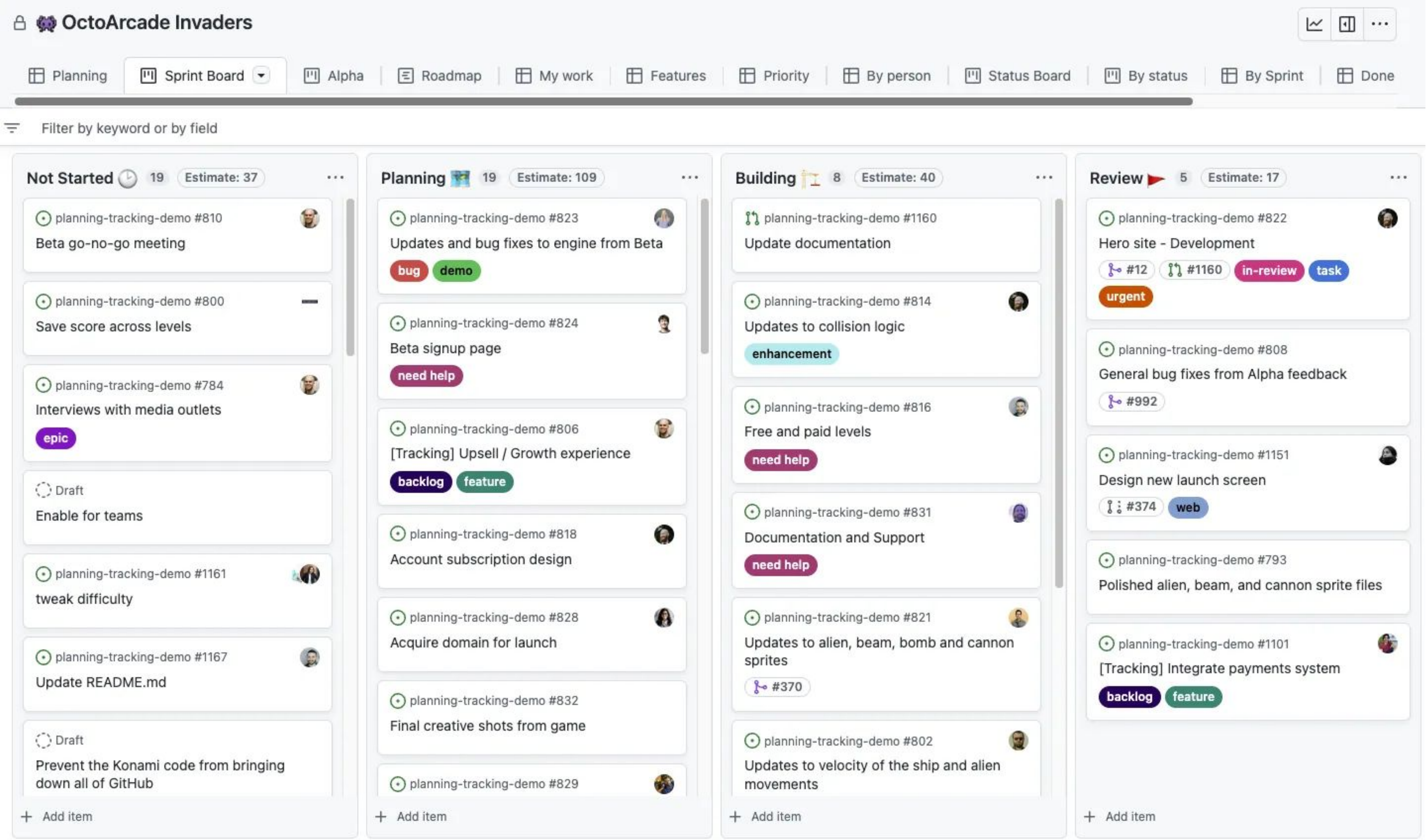
Pull Request

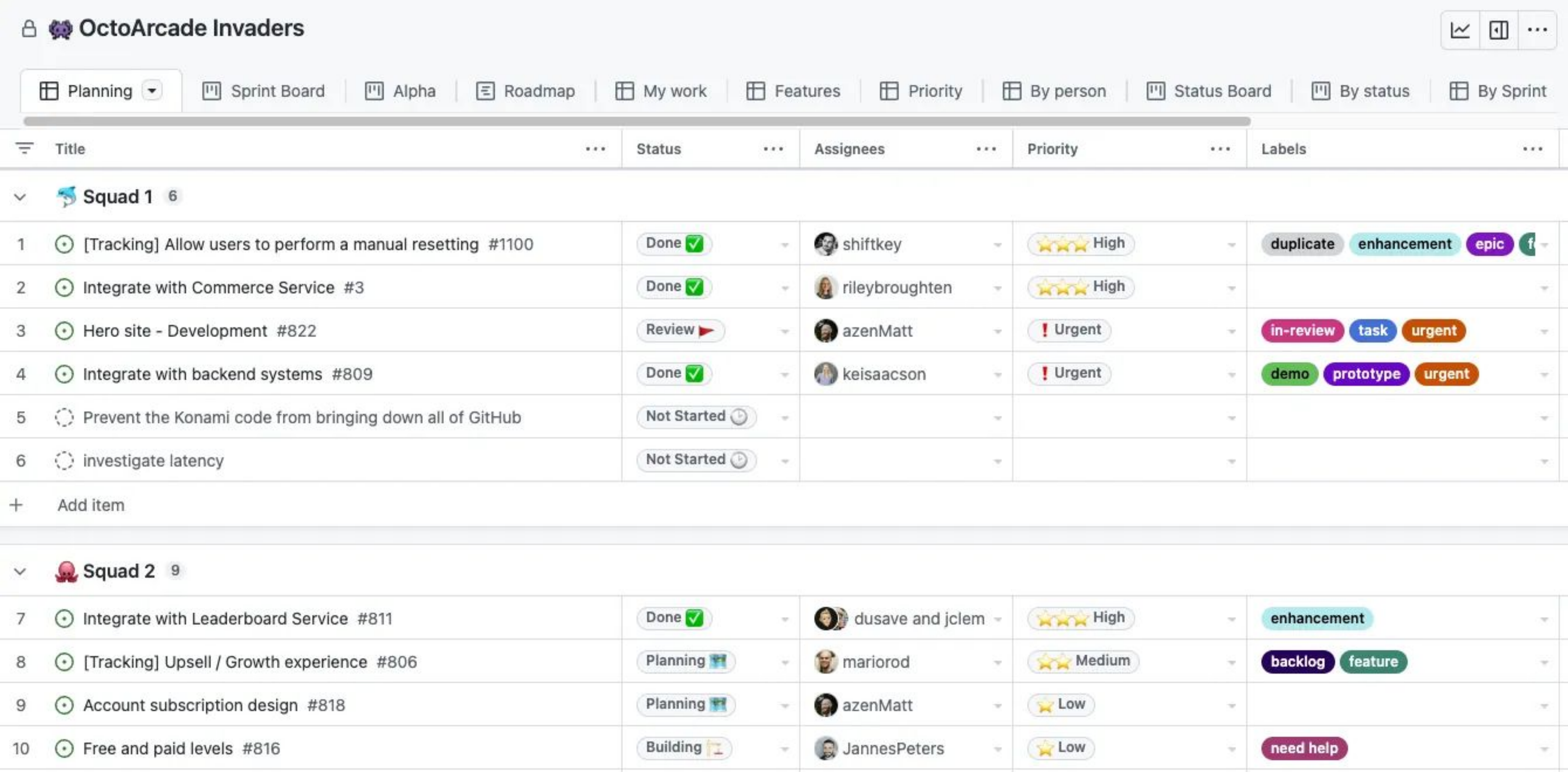
Asks the maintainers of the original repository to take a look at and hopefully integrate your code changes into their repository.



GitHub Issues are used to track tasks, enhancements, and bugs for a project, allowing team members to discuss and manage work collaboratively.

GitHub Projects is a tool to organize and track the progress of issues and pull requests in a board view, helping teams visualize workflows and prioritize tasks.





Data wrangling

Data wrangling deals with several functionalities:

1. Data exploration: In this process, the data is studied, analyzed and understood by visualizing representations of data.

2. Dealing with missing values: Most of the datasets having a vast amount of data contain missing values of NaN, they are needed to be taken care of by replacing them with mean, mode, the most frequent value of the column or simply by dropping the row having a NaN value.

Data wrangling

Data wrangling deals with several functionalities:

1. Data exploration

2. Dealing with missing values

3. Reshaping data: In this process, data is manipulated according to the requirements, where new data can be added or pre-existing data can be modified.

4. Filtering data: Some times datasets are comprised of unwanted rows or columns which are required to be removed or filtered

pandas and numpy

import pandas as pd

import numpy as np

pandas	numpy
When we have to work on Tabular data, we prefer the pandas module.	When we have to work on Numerical data, we prefer the NumPy module.
Pandas have a 2D table object called DataFrame.	Numpy is capable of providing multi-dimensional arrays.
The powerful tools of pandas are DataFrame and Series.	the powerful tool of NumPy is Arrays.
Pandas consume more memory.	Numpy is memory efficient.
Indexing of the Pandas series is very slow as compared to Numpy arrays.	Indexing of Numpy arrays is very fast.

https://www.geeksforgeeks.org/difference-between-pandas-vs-numpy/

pandas operation

read csv files into a pandas df: pd.read_csv("link")

Dataset

1. kaggle
https://www.kaggle.com/datasets

2. UCI Machine Learning Repository
https://archive.ics.uci.edu/datasets

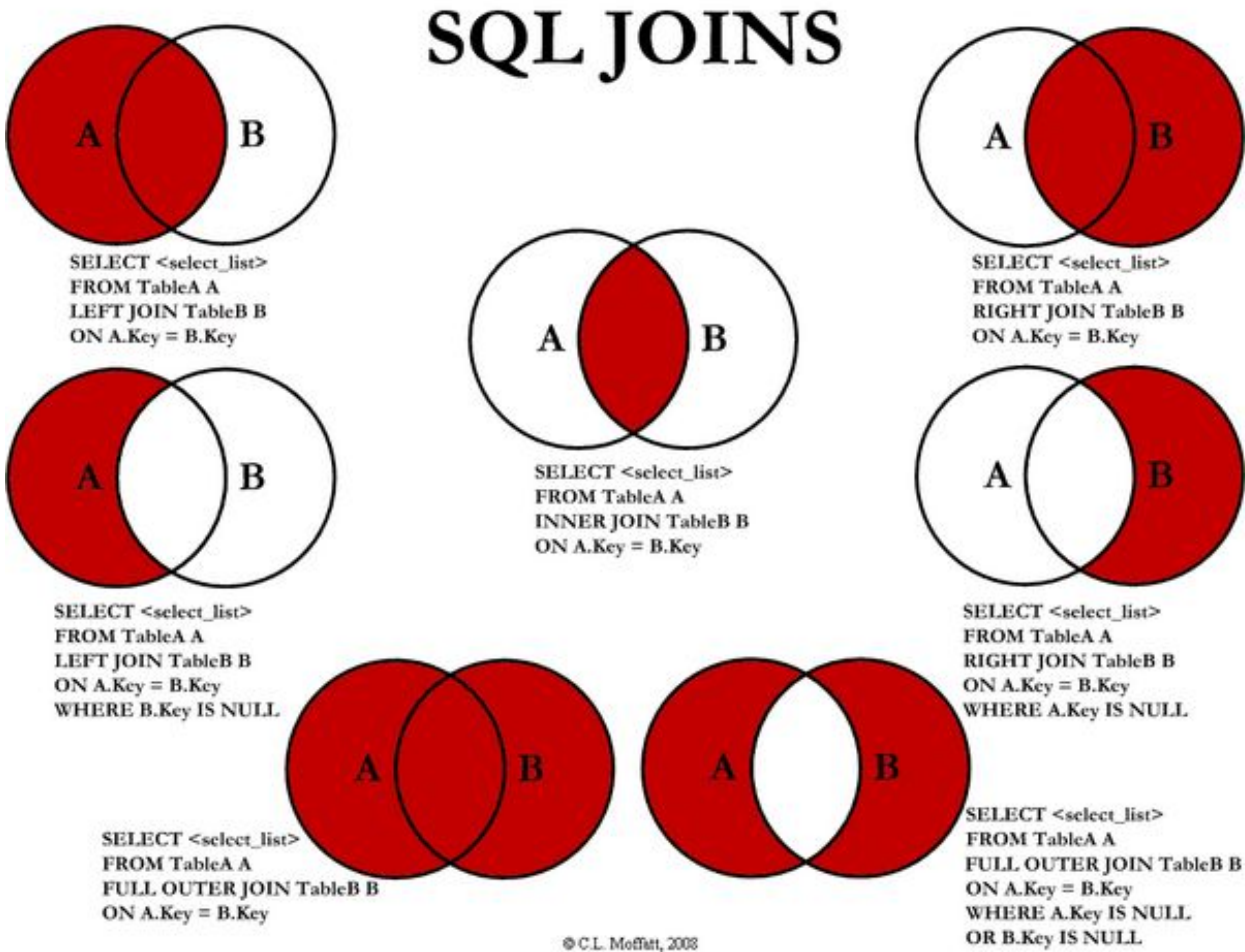
3. Google Dataset Search
https://datasetsearch.research.google.com/

4. Github or other resources

Best practices when working with large data sets

Be careful when joining, aggregating, sorting, or filtering your dataset.

Make sure to limit the dataset to only columns and rows needed.



Your time to …

- Start looking for datasets for your final projects
- Work on D2